

80386 Operating Modes (Real, Protected & Virtual 8086) — MPCA Module 5

Definition: The 80386 microprocessor supports **three distinct operating modes**: **Real Address Mode** (16-bit 8086 compatible), **Protected Virtual Address Mode** (32-bit multitasking protected environment), and **Virtual 8086 (V86) Mode** (running real-mode 8086 software inside protected mode).

1. Detailed Comparison of 80386 Operating Modes

80386 OPERATING MODES		
1. REAL MODE:	2. PROTECTED MODE:	3. VIRTUAL 8086:
- Default upon power-on	- PE bit = 1 in CR0	- VM bit = 1
- Exact 8086 clone	- 32-bit Flat/Segmented	- Run 8086 apps
- 1 MB Physical RAM	- 4 GB RAM / 64 TB Virtual	as task under
- No memory protection	- Hardware Privilege 0-3	multitasking OS

2. Deep Dive into Each Mode

1. Real Address Mode (Real Mode)

- Processor boots in this mode upon power-up/reset.
- Behaves as a high-speed 8086 processor:
- Base Address = $\text{Segment} * 16 + \text{Offset}$
- Max addressable memory = 1 MB (plus 64KB High Memory Area HMA).
- No memory protection or privilege rings (single buggy application can crash the entire system).

2. Protected Virtual Address Mode (Protected Mode)

- Activated by setting **PE = 1** in Register CR0.
- **Features:**
 - Full 32-bit 4 GB address space with optional Paging.
 - **4 Privilege Levels (Rings 0 to 3):**
 - Ring 0 (Kernel / OS Core) - Highest Privilege.
 - Ring 1 (Device Drivers).
 - Ring 2 (OS Extensions / Middleware).
 - Ring 3 (User Applications) - Lowest Privilege.
- Segment registers hold **16-bit Selectors** pointing to 8-byte **Descriptors** in Global Descriptor Table (GDT) or Local Descriptor Table (LDT).

3. Virtual 8086 Mode (V86 Mode)

- Sub-mode of Protected Mode enabled by setting **VM = 1** in **EFLAGS register**.
- Allows the multitasking OS to execute multiple legacy DOS / 8086 programs as isolated Ring 3 user tasks with hardware memory protection!

3. Operating Mode Comparison Table

Property	Real Mode	Protected Mode	Virtual 8086 Mode
Address Space	1 MB	4 GB Physical / 64 TB Virtual	1 MB per V86 task
Memory Protection	None	Full 4-Level Rings (0-3)	Protected under Ring 3
Multitasking	Not supported in hardware	Hardware Task Switching (TSS)	Multitasked by OS kernel
Paging Support	Disabled	Full 4 KB Paging (PG bit)	Supported via Paging MMU

4. Quick Recall Flow

Power-on -> Real Mode (8086 clone 1MB) -> Set PE=1 in CR0 -> Protected Mode (4GB, Rings 0-3) -> Set VM=1 -> Virtual 8086 Mode